

STRATASEARCH

ADVANCED SEARCH & TEXT ANALYTICS LABORATORY

An Interactive Platform for Visualizing, Benchmarking, and Comparing Advanced Text Processing Algorithms

PROJECT TEAM

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PROJECT GUIDE

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KL UNIVERSITY

StrataSearch Bridges the Gap Between Theoretical Learning & Practical Implementation

Motivation

Conventional search systems only display results. StrataSearch explains the internal working of algorithms through interactive visualization and benchmarking.

Objective

To build an interactive text analytics platform that helps students understand practical applications of Data Structures and Algorithms.

Key Features

Platform capabilities & core tools

Smart Search Engine

Algorithm Visualizer

Benchmark Dashboard

Doc Similarity Analyzer

Citation Graph

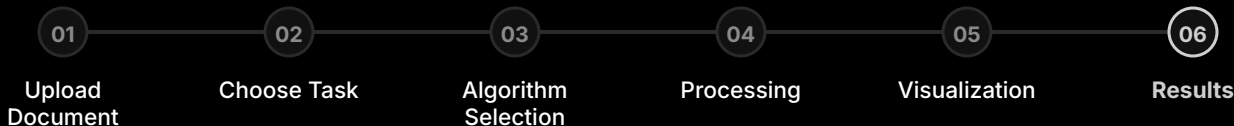
Complexity Analyzer

Dataset Manager

Interactive Platform

Workflow

Document analysis & processing pipeline



Functional Responsibilities of Team Members



Member 01

Search Engine Core

Responsibilities

- Develops Search Engine Core
- Implements KMP & Rabin–Karp
- Pattern & Phrase Search
- Search Ranking Logic
- Search API Integration



Member 02

Advanced Text Analysis Core

Responsibilities

- Implements Z Algorithm
- Implements Aho–Corasick
- Develops Suffix Array & LCP
- Document Similarity Analysis
- Duplicate Detection
- Complexity Analyzer



Member 03

Analytics, Visualization & Platform Core

Responsibilities

- Algorithm Visualizer
- Benchmark Dashboard
- Research Paper Recommendation
- Citation Graph
- Dataset Manager

Collaborative Focus

Shared ownership of documentation, Git/GitHub version control, end-to-end integration, and final project presentation.

Expected Outcomes of StrataSearch

Delivering a scalable platform for exploring and benchmarking advanced text processing algorithms.

Algorithms



Visualization



Benchmarking



Insights

01 Practical Learning

Interactive learning and practical implementation of advanced DSA concepts.

02 Search & Similarity

Fast document searching and document similarity analysis.

03 Benchmarking

Comprehensive algorithm benchmarking comparing runtime, memory, and execution time.

04 Educational Support

Performance visualization and educational support for students.

Conclusion: The platform demonstrates how multiple algorithms work together to solve modern text analytics problems efficiently based on task and dataset characteristics.

REAL-WORLD IMPACT

Applying advanced string processing algorithms to solve real-world text analytics challenges.

ADDITIONAL APPLICATIONS

Search Engines • Data Analytics
AI Research

Engineering Education

Visualizing complex structures and providing educational support for computer science students.

Research & Libraries

Benchmarking novel text algorithms and analyzing academic databases.

Plagiarism Detection

Finding structural similarity and duplicate detection across large document corpuses.

Enterprise Knowledge

Efficiently navigating unstructured text assets and enterprise search.

Software Engineering

Training engineers in highly optimized search solutions.

Collaborative Version Control & Project Architecture

Repository Information

Name: A-7_DSA_G-7

Link: [sashank321/A-7_DSA-3_G-7](https://github.com/sashank321/A-7_DSA-3_G-7)



Development Workflow

Utilizing **Git and GitHub** for collaborative version control, issue tracking, documentation, and seamless code management across the development team.

Project Folder Structure

- 📁 **frontend/** React, Tailwind CSS, Chart.js
- 📁 **backend/** Java, Spring Boot
- 📁 **database/** PostgreSQL
- 📁 **algorithms/**
- 📁 **docs/**
- 📁 **assets/**

StrataSearch Supporting References

- **Introduction to Algorithms** — CLRS
- **Algorithms** — Jeff Erickson
- **Algorithm Design** — Kleinberg & Tardos
- **Technical Documentation:** Java, Spring Boot, React, PostgreSQL, and Graphviz.

THANK YOU. QUESTIONS?